HERMES SANDALS

SMALL WINGS. DIVINE RANGE.

A courier myth rebuilt as material, energy and proxy choices.

STEP ONE

ONE PAIR. ONE DIVINE MISSION.

The model treats Hermes' talaria as engineered footwear: a leather pair with bronze wing frames, feather vanes and a thin gold finish.

FUNCTIONAL UNIT

One pair

Winged talaria enabling one courier mission.

REFERENCE FLOW

0.682 kg

Pair mass excluding reusable narrative packaging.

TIMEFRAME

Single build

Cradle-to-delivered-gate; no operational fuel.

ATTRIBUTE	MODELED VALUE
MASS	0.682 kg product
DESIGN	leather + bronze + feathers + gold
CONTEXT	Greek myth reconstructed with 2026 factors
CUTOFF	wear, repair and divine propulsion excluded

The service is not flight itself; it is the delivered artifact that makes flight possible.

STEP ONE

A SMALL OBJECT. A LONG TRAIL.

Included stages stop at delivery. The mythic use phase is modeled explicitly as zero direct energy, not hidden inside the result.

INCLUDED	TREATMENT
Leather, metal, feathers	mass-based material flows
Gold leaf	external gold proxy
Workshop energy	UK electricity consumption
Packaging	wood and paper flows
Road and sea freight	tonne-km logistics

EXCLUDED	REASON
Divine propulsion	no measurable fuel
Olympian maintenance	no reference data
End-of-life	immortal artifact assumption
Avoided travel	no comparative assertion

CUTOFF EVENT

The system ends when the packed sandals reach the mythic delivery gate. Use-phase flight is stated as zero direct emissions.

The boundary keeps the myth visible and prevents flight magic from masking material impacts.

STEP TWO

THE TALARIA BECOME HARDWARE.

The canonical object is fixed before calculation: paired sandals, ankle straps, bronze-gold wing frames, pale feather vanes and small emerald inlays.

PROPERTY	VALUE
Sole length	270 mm
Sole width	95 mm
Wing height	310 mm
Product mass	0.682 kg
Shipped mass	0.862 kg
Gold leaf	2.0 g

GEOMETRY RULE

The sandals are not solid gold. Gold is modeled as a thin decorative layer over metal, because solid gold footwear would be physically implausible and structurally excessive.

The design is extravagant, but its mass remains footwear-scale.

STEP TWO

THE INVENTORY HAS A HOTSPOT BEFORE THE RESULT.

The largest mass is leather, but the highest intensity belongs to gold. Both are kept visible before the baseline is shown.

FLOW	QUANTITY	FACTOR BASIS	NOTE
Leather body	0.440 kg	DEFRA clothing proxy	main structure
Gold leaf	0.002 kg	S&P proxy	ornamental finish
Feather/textile	0.060 kg	DEFRA clothing proxy	wing vanes
Bronze/brass	0.180 kg	DEFRA metals	frame and buckles
Packaging	0.180 kg	DEFRA wood/paper	box and wrap
Energy + water	4.0 kWh; 0.02 m3	DEFRA utilities	workshop
Freight	0.793 t-km	DEFRA freight	road + sea

EXPECTED HOTSPOT

The two grams of gold are expected to dominate the result despite their near-zero mass share.

The inventory is ordered by expected impact, not by mass.

STEP TWO

FIVE STEPS, ONE ARTIFACT.

The production model compresses the myth into a workshop sequence that can be counted without pretending to be a factory audit.

01 **Cut and stitch leather**

02 **Forge bronze wing frame**

03 **Mount feather vanes**

04 **Apply gold leaf**

05 **Pack for delivery**

The process sequence supports the inventory; it does not add unmodeled stages.

STEP TWO

THE ROAD IS SMALLER THAN THE GILDING.

Logistics are counted as tonne-kilometres. A distance alone is not a freight activity, so shipped mass remains explicit.

WORKSHOP

4.0 kWh

UK electricity + T&D + WTT.

WATER

0.02 m3

Supply + treatment.

ROAD

120 km

0.103 t-km.

SEA

800 km

0.690 t-km.

LOGISTICS CONTRIBUTION

0.031 kg CO2e

Less than 0.1% of the modeled baseline.

For this object, logistics are measurable but not material.

STEP THREE

THE FACTORS DEFINE THE ANSWER.

DEFRA 2026 is used first. Where it lacks a representative factor, the model uses an external proxy and names the bias.

ID	FACTOR	VALUE	UNIT	SOURCE
S1	Clothing primary	22.310	kg/kg	DEFRA proxy
S2	Metals primary	3.822	kg/kg	DEFRA
S5	UK electricity use	0.1475	kg/kWh	DEFRA sum
S6	HGV freight	0.1272	kg/t-km	DEFRA + WTT
S7	Container ship	0.0198	kg/t-km	DEFRA + WTT
S9	Primary gold	25,463	kg/kg	S&P proxy

PROXY DISCLOSURE

Leather uses DEFRA clothing as the baseline proxy; gold uses an external mining-intensity proxy because DEFRA has no representative gold factor.

The model is transparent where it is precise, and transparent where it is not.

STEP FOUR

TWO GRAMS BEND THE RESULT.

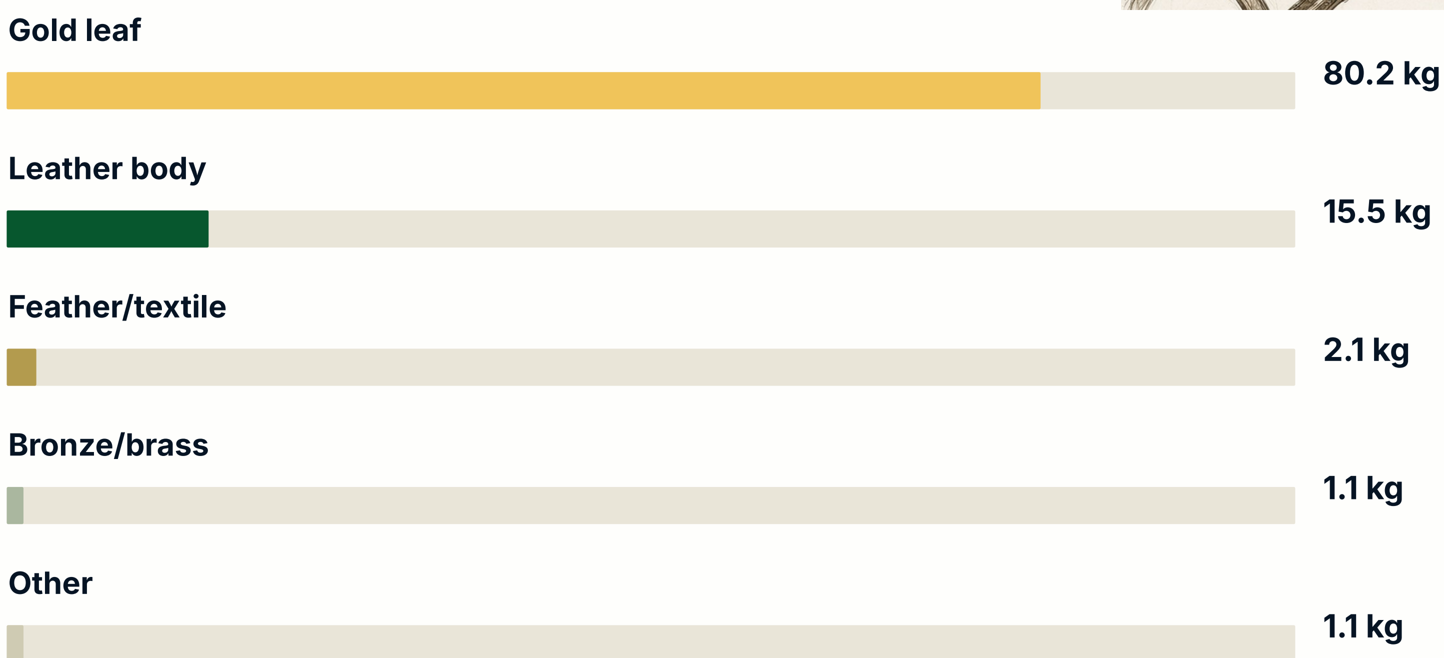
The baseline footprint is controlled by the decorative gold layer, not by the visible leather mass.

BASELINE RESULT

63.5

kg CO2e per pair

Cradle-to-delivered-gate, single pair reference flow.



HOTSPOT

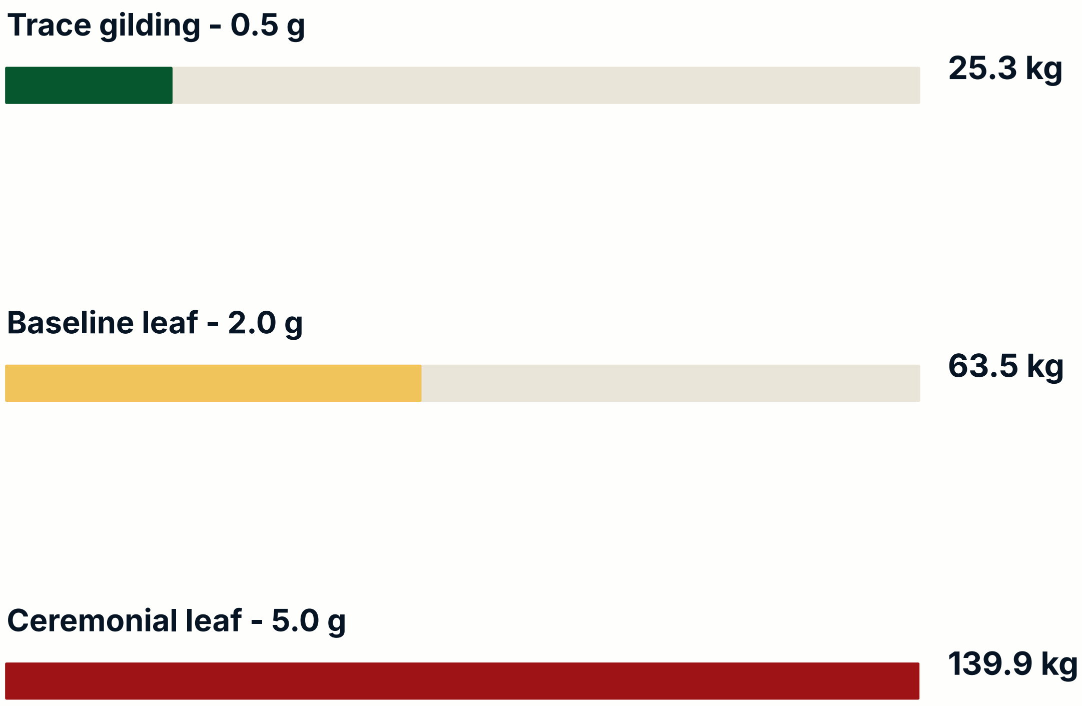
Gold contributes 80.2% of the baseline with only 0.3% of product mass.

A tiny high-intensity input outweighs nearly everything else.

STEP FIVE

GILDING IS THE CONTROL KNOB.

Sensitivity A changes one physical parameter only: gold leaf mass. All other flows remain fixed.



SENSITIVITY RULE

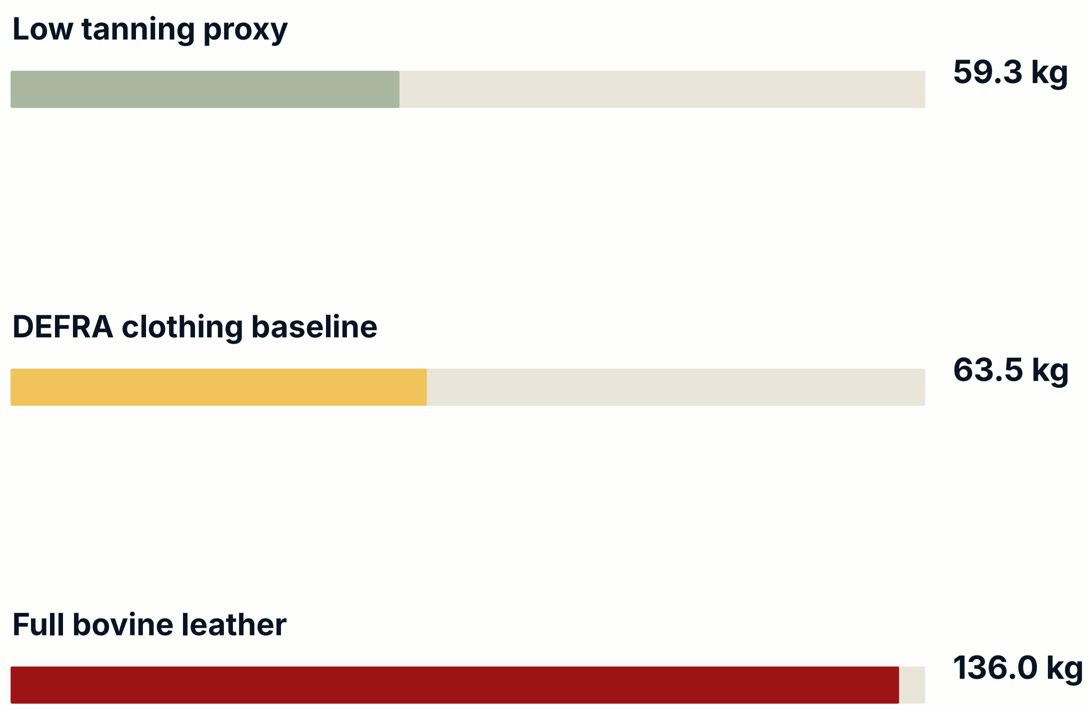
The baseline remains 63.5 kg CO2e. Increasing gold from 2 g to 5 g more than doubles the total footprint.

In this model, shine is a physical parameter with systemic consequence.

STEP FIVE

THE LEATHER PROXY MATTERS SECOND.

Sensitivity B changes the leather emission factor. Mass, gold, energy and logistics stay fixed.



CASE	LEATHER EF	TOTAL	USE
Low	12.9 kg/kg	59.3 kg	literature lower bound
Baseline	22.31 kg/kg	63.5 kg	DEFRA proxy
High	187.1 kg/kg	136.0 kg	full bovine leather benchmark

Even the high leather case does not erase the gold hotspot logic.

STEP SIX

FINDING. WHAT THE MODEL TELLS US.

Four conclusions are drawn only from the modeled values already shown.

01 GOLD DOMINATES

2.0 g of gold leaf accounts for 80.2% of the baseline.

02 LEATHER IS STILL VISIBLE

The DEFRA clothing proxy contributes 9.82 kg CO2e.

03 FREIGHT IS COUNTED

Road and sea logistics add only 0.031 kg CO2e.

04 MAGIC IS NOT A FACTOR

Use-phase propulsion is explicitly set to zero direct emissions.

“The smallest shine can carry the longest shadow.”

The conclusion is a material claim, not a mythic compliment.

MYTHS & LEGENDS

Impossible objects, reconstructed as systems.

HERMES SANDALS

A small golden detail can define the footprint of flight.